

TF-IDF and Logistic Regression as a Baseline Model:
A Methodological Analysis for Multi-Label Text Classification
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Abstract

This paper provides an academic analysis of the theoretical foundations and methodological significance of selecting TF-IDF (Term Frequency-Inverse Document Frequency) combined with Logistic Regression as the baseline model in the RAISE-26 multi-label text classification task. Based on a corpus of 10,500 AI-related news headlines annotated with 12 behavioral impact categories, this study systematically examines six critical dimensions: (a) methodological significance of baseline selection, (b) text representation theory, (c) mathematical properties of linear classifiers, (d) multi-label learning strategies, (e) model interpretability, and (f) computational efficiency. Results demonstrate that the baseline achieves Micro-F1 of .943 and Macro-F1 of .933, establishing a meaningful performance floor for evaluating more complex deep learning approaches. The analysis contributes to methodological best practices in applied machine learning research.

Keywords: baseline model, TF-IDF, logistic regression, multi-label classification, text classification, machine learning methodology

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Methodological Significance of Baseline Selection

The Role of Baselines in Machine Learning Research

In machine learning research, baseline model selection constitutes not merely a technical decision but a fundamental methodological consideration. A baseline serves as the reference point against which all subsequent improvements are evaluated (Armstrong, Moffat, Webber, & Zobel, 2009). Without an appropriately chosen baseline, it becomes impossible to assess whether complex models provide genuine empirical value or merely introduce unnecessary computational overhead.

The significance of baselines extends beyond simple performance comparison. As noted by Lipton (2018), the absence of strong baselines has led to inflated claims of improvement in numerous machine learning studies. This observation underscores the critical importance of establishing well-justified reference points before exploring sophisticated alternatives.

Criteria for Baseline Selection

According to Armstrong et al. (2009), a well-chosen baseline should satisfy four essential criteria:

Representativeness. The baseline should represent standard practice in the task domain, reflecting what a competent practitioner would reasonably attempt as an initial approach. This criterion ensures that the baseline captures accepted methodological norms rather than artificially weak alternatives.

Reproducibility. The baseline should be implemented using mature, well-documented, open-source tools that other researchers can easily replicate. This criterion facilitates verification and extension of reported results.

Comparability. The baseline should facilitate horizontal comparison with other research in the field, enabling meaningful benchmarking across studies. This criterion situates individual contributions within the broader research landscape.

Lower bound property. The baseline should provide a meaningful performance floor that more complex models must exceed to justify their additional complexity. This criterion establishes the threshold for demonstrating genuine improvement.

Justification for TF-IDF and Logistic Regression

Representativeness. The combination of TF-IDF and Logistic Regression has served as the de facto standard for text classification tasks for over two decades (Joachims, 1998). From spam filtering to sentiment analysis, this approach represents the canonical first attempt that practitioners typically employ before exploring more complex alternatives. Its widespread adoption in both academic research and industrial applications establishes its representativeness as a baseline choice.

Reproducibility. The scikit-learn library (Pedregosa et al., 2011) provides standardized, extensively tested implementations of both TF-IDF Vectorizer and Logistic Regression. These implementations are thoroughly documented, widely validated, and produce consistent results across different computational environments. Any researcher can reproduce the baseline with minimal technical barriers.

Comparability. Nearly all text classification publications in major natural language processing venues (e.g., ACL, EMNLP, NAACL) report TF-IDF and Logistic Regression as a baseline. This ubiquity enables direct comparison of results across hundreds of prior studies, situating contributions within established benchmarks.

Lower bound property. As demonstrated in the experimental evaluation, this baseline achieves Micro-F1 of .943 and Macro-F1 of .933 on the test set. These results establish a meaningful performance floor: the DistilBERT model must demonstrably exceed these metrics to justify its additional computational cost and reduced interpretability.

The Simple Baseline Paradox

Recent research in natural language processing has identified an important phenomenon: on many tasks, the performance gap between simple baseline models and complex deep learning approaches is substantially smaller than commonly assumed (Lipton, 2018). The present study supports this observation:

Table 1

Performance Comparison Between Baseline and Deep Learning Models

Model	Micro-F1	Macro-F1	Training Time
TF-IDF + Logistic Regression	.943	.933	~5 min
DistilBERT	~.950	~.940	30-60 min

The marginal improvement of approximately 1-2% in F1 scores is accompanied by a 6-12x increase in training time and substantially reduced interpretability. This trade-off analysis is rendered possible only through the establishment of a strong, well-justified baseline.

TF-IDF Text Representation: Theoretical Foundation

Information Retrieval Theory

Term Frequency-Inverse Document Frequency (TF-IDF) originates from classical information retrieval theory, with its foundational concepts established through the pioneering contributions of Luhn (1957) and Sparck Jones (1972). The fundamental insight underlying TF-IDF is that the importance of a term in a document should be weighted by both its local frequency within the document and its global rarity across the corpus.

The TF-IDF weight for term t in document d within corpus D is formally defined as:

$$TF\text{-}IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

where $TF(t, d)$ represents the term frequency of t in document d , and $IDF(t, D) = \log(|D| / |\{d \in D: t \in d\}|)$ represents the inverse document frequency, calculated as the logarithm of the total number of documents divided by the count of documents containing term t .

Suitability for News Headline Classification

Zipf's Law applicability. News headlines conform to the Zipf distribution characteristic of natural language, wherein a small number of high-frequency words convey general semantics while medium-to-low frequency terms carry discriminative information (Zipf, 1949). TF-IDF effectively attenuates the influence of high-frequency stopwords through IDF weighting, thereby highlighting discriminative vocabulary that distinguishes between behavioral categories.

Bag-of-words assumption. For short texts such as news headlines, the importance of word order information is diminished relative to longer documents. The core semantics of headlines are predominantly determined by key vocabulary rather than complex syntactic structures. This

characteristic renders the bag-of-words model, which disregards word order, a reasonable simplification for this specific task domain.

N-gram semantic enhancement. The `ngram_range=(1, 2)` configuration employed in this study introduces bigram features, capturing phrase-level semantic units such as "AI assistant," "job loss," and "privacy concern." This enhancement improves semantic expressiveness while preserving the computational advantages of sparse representation.

Configuration Parameters

Table 2

TF-IDF Vectorizer Configuration and Rationale

Parameter	Value	Rationale
<code>ngram_range</code>	(1, 2)	Balances vocabulary granularity with feature space expansion
<code>min_df</code>	2	Filters noise from singleton terms with limited discriminative value
<code>max_features</code>	60,000	Controls dimensionality based on information-theoretic principles
<code>stop_words</code>	english	Removes function words to focus on content word discrimination

Logistic Regression: Theoretical Analysis

Mathematical Foundation

Logistic Regression is a discriminative probabilistic model that directly estimates the posterior probability $P(Y|X)$. In contrast to generative models that model the joint distribution $P(X, Y)$, discriminative models focus on learning the decision boundary directly, which is often more efficient for classification tasks (Ng & Jordan, 2002).

For binary classification, the probability of the positive class is modeled as:

$$P(Y = 1|X) = \sigma(w'X + b) = 1 / (1 + \exp(-(w'X + b)))$$

where $\sigma(\cdot)$ denotes the sigmoid function, w represents the weight vector, and b is the bias term. The sigmoid function maps real-valued inputs to the interval $(0, 1)$, providing a probabilistic interpretation of classifier output.

Suitability for High-Dimensional Sparse Data

Computational efficiency. TF-IDF produces high-dimensional sparse feature vectors (dimensionality exceeding 60,000 in this project). As a linear model, Logistic Regression exhibits computational complexity proportional to the number of non-zero features rather than total features. This property provides substantial computational advantages when processing sparse text representations (Joachims, 1998).

Probabilistic output. Unlike Support Vector Machines that produce hard boundary decisions, Logistic Regression outputs probability values. This enables threshold adjustment (e.g., the .50 threshold employed in this study) in multi-label scenarios to balance precision and recall, providing flexibility for tuning model behavior to specific application requirements.

Regularization. The L2 regularization incorporated in the model configuration (default behavior of the liblinear solver) effectively controls overfitting risk in high-dimensional feature spaces. This approach conforms to the structural risk minimization principle in statistical learning theory (Vapnik, 1995).

Classifier Configuration

Table 3

Logistic Regression Configuration and Rationale

Parameter	Rationale
class_weight="balanced"	Addresses label imbalance through automatic loss weight adjustment; equivalent to cost-sensitive learning
solver="liblinear"	Employs coordinate descent optimization; suitable for L1/L2 regularized problems (Fan, Chang, Hsieh, Wang, & Lin, 2008)
max_iter=2000	Ensures convergence; prevents underfitting from premature termination

One-vs-Rest Multi-Label Learning Strategy

Problem Transformation Approach

The core challenge of multi-label learning lies in modeling label correlations. The One-vs-Rest (OvR) strategy adopts the problem transformation approach, decomposing the multi-label problem into multiple independent binary classification problems (Tsoumakas & Katakis, 2007).

For K labels, the OvR strategy trains K independent binary classifiers:

$$f_k(x) = P(y_k = 1 | x), \text{ for } k = 1, 2, \dots, K$$

The final prediction applies a threshold τ (set to .50 in this study): $\hat{y}_k = 1$ if $f_k(x) \geq \tau$, otherwise $\hat{y}_k = 0$.

Advantages

The OvR strategy offers several methodological advantages:

Conceptual simplicity. The approach is straightforward to understand, implement, and debug, reducing the risk of implementation errors.

Parallelizability. Each label classifier can be trained independently, enabling efficient parallel computation.

Interpretability. Decisions for each label can be explained independently, facilitating error analysis and model refinement.

Flexibility. Different thresholds or even different model architectures can be employed for different labels based on their characteristics.

Limitations

Despite its advantages, the OvR strategy has recognized limitations:

Label independence assumption. OvR assumes conditional label independence, potentially failing to capture co-occurrence patterns between categories such as "Work, Jobs and Economy" and "Learning, Knowledge and Education."

Class imbalance amplification. Each binary classifier faces more severe class imbalance than the original multi-label problem, as each label typically applies to a minority of instances.

Empirical Justification

Despite theoretical limitations, the experimental results demonstrate that the OvR strategy achieves strong performance (Micro-F1 = .943, Macro-F1 = .933) on this dataset. This finding suggests that either label correlations are relatively weak in this domain, or the discriminative capacity of TF-IDF features has implicitly captured sufficient inter-label relationships. The close alignment between Micro-F1 and Macro-F1 further indicates balanced performance across all 12 behavioral categories.

Interpretability Analysis

Transparency of Linear Models

A significant advantage of combining TF-IDF with Logistic Regression lies in its complete interpretability, which is particularly valuable under the current Explainable AI (XAI) paradigm (Lipton, 2018). The increasing demand for model transparency in research and regulatory contexts renders interpretable baselines especially important.

Levels of Interpretability

Feature level. TF-IDF weights directly correspond to interpretable vocabulary features, enabling researchers to examine which terms contribute most strongly to classifying specific categories.

Model level. The weight coefficients w_i in Logistic Regression directly represent the contribution direction and magnitude of feature i to classification decisions. Positive weights indicate positive association with the target class; negative weights indicate negative association.

Prediction level. Individual predictions can be decomposed into feature contributions through the linear combination $\sum (w_i \times x_i)$, achieving local interpretability for any specific classification decision.

Research Value

For the RAISE-26 research project investigating AI's behavioral impact on humans, interpretability provides value:

Hypothesis validation. Researchers can verify whether the model has learned semantically reasonable feature associations (e.g., whether "automation" strongly associates with "Work, Jobs and Economy").

Error diagnosis. Analysis of feature patterns in misclassified cases can inform improvements to model design or data quality.

Knowledge discovery. The model can reveal latent patterns in data, such as unexpected vocabulary associations with specific behavioral categories.

Computational Efficiency Analysis

Time Complexity

The training complexity of TF-IDF combined with Logistic Regression can be expressed as:

$$O(n \times d \times K \times I)$$

where n represents the number of samples (10,500), d denotes the average number of non-zero features (substantially less than 60,000 due to sparsity), K indicates the number of labels (12), and I represent the maximum number of optimization iterations (2,000).

Comparison with Deep Learning

Table 4

Computational Requirements Comparison

Model	Training Time	Inference Speed	Hardware
TF-IDF + Logistic Regression	~5 minutes	Very fast	CPU only
DistilBERT	30-60 minutes	Moderate	GPU recommended

Practical Implications

The computational efficiency of the baseline enables several practical research activities:

Rapid prototyping. Data processing pipelines can be validated within minutes, accelerating the experimental cycle.

Hyperparameter optimization. The low training cost enables exploration of larger hyperparameter search spaces.

Cross-validation. Multi-fold cross-validation can be performed without prohibitive time costs.

Ablation studies. The impact of different feature configurations can be rapidly assessed.

Role as Benchmark for Deep Learning Comparison

Establishing Reference Points

In the project pipeline, the TF-IDF and Logistic Regression baseline serves as the critical reference point for evaluating the DistilBERT deep learning model. This comparison framework enables researchers to quantify the empirical value added by contextual embeddings and transformer architectures (Devlin, Chang, Lee, & Toutanova, 2019).

Complementary Characteristics

Table 5

Comparative Characteristics of Baseline and Deep Learning Models

Dimension	TF-IDF + Logistic Regression	DistilBERT
Feature representation	Discrete vocabulary	Continuous embeddings
Context modeling	None (bag-of-words)	Bidirectional (self-attention)
Out-of-vocabulary handling	Limited	Robust (subword tokenization)
Training data requirements	Lower	Higher
Interpretability	Full transparency	Limited (black box)

Model Selection Framework

Based on the analysis presented in this study, the following decision framework is proposed for practitioners:

Rapid prototyping or exploratory analysis: TF-IDF and Logistic Regression is recommended due to fast iteration cycles and immediate interpretability.

Production systems prioritizing performance: DistilBERT may be preferred when the marginal performance improvement justifies increased computational costs.

Interpretability requirements: TF-IDF and Logistic Regression is strongly recommended when model decisions must be explained to stakeholders.

Computational resource constraints: TF-IDF and Logistic Regression enables effective modeling without GPU infrastructure.

Conclusion

In conclusion, the TF-IDF and Logistic Regression baseline achieves Micro-F1 of .943 and Macro-F1 of .933 on the RAISE-26 multi-label classification task, establishing a strong methodological foundation for the research project. The 93%+ F1 performance, combined with complete interpretability, minimal computational requirements (approximately 5 minutes training on CPU), and theoretical grounding in established information retrieval and statistical learning frameworks, validates this approach as an appropriate baseline for evaluating more complex deep learning alternatives. The marginal 1-2% improvement offered by DistilBERT must be weighed against 6-12x increased computational cost and substantially reduced interpretability.

This paper has presented a comprehensive academic analysis of the selection of TF-IDF combined with Logistic Regression as the baseline model for multi-label text classification in the RAISE-26 research project. The analysis demonstrates that this choice satisfies established methodological criteria for baseline selection: representativeness as a decades-long standard approach, reproducibility through standardized open-source implementations, comparability enabling cross-study benchmarking, and the lower bound property establishing meaningful performance thresholds.

The theoretical foundations of both TF-IDF (grounded in information retrieval theory) and Logistic Regression (grounded in statistical learning theory) are well-established and appropriate for the task characteristics of this study. The One-vs-Rest multi-label strategy, while theoretically limited in capturing label correlations, demonstrates strong empirical performance that justifies its use as a baseline approach.

The interpretability and computational efficiency of the baseline model provide substantial practical value for research activities, enabling rapid hypothesis testing, error diagnosis, and knowledge discovery. Finally, the baseline serves its essential function of establishing a meaningful reference point for evaluating more complex deep learning approaches, enabling rigorous assessment of the trade-offs between performance, interpretability, and computational cost.

Future research directions include exploration of label correlation modeling through methods such as classifier chains (Read, Pfahringer, Holmes, & Frank, 2011), threshold optimization for individual labels, and systematic evaluation of ensemble approaches combining baseline and deep learning predictions.

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Appendix

Prompts by Cell Number:

Stage	Cells	Key Components
1. Environment Setup	4, 6, 8	Dependencies, imports, random seeds, path config
2. Data Loading	13, 15, 17, 19	CSV loading, label parsing, distribution analysis
3. Preprocessing	23, 25, 28	Text cleaning, rare label handling, label mappings
4. Encoding & Splitting	32, 34, 36	Multi-hot encoding, stratified sampling
5. Baseline Model	40, 42, 44, 47	TF-IDF, LogReg, OvR classifier, evaluation
6. DistilBERT	51, 53, 55, 57, 59, 61	Dataset class, model, trainer, training loop
7. Evaluation	65, 68, 71	Test evaluation, comparison table, confusion matrices
8. Interpretability	75, 77, 80	Feature extraction, keyword analysis, error analysis
9. Topic Modeling	84, 86, 88	NMF, topic keywords, distribution visualization
10. Dataset C Analysis	92-116	LLM comparison, statistical tests, dashboards
11. Quant Analysis	122-144	Market data, GARCH, Granger causality, correlations

1. Stage 1: Environment Setup

Notebook Reference: Cells 4, 6, 8

Purpose: Establish a fully reproducible computational environment for the entire pipeline, ensuring consistent results across different runs and platforms.

AI Prompt:

Design a reproducible Python environment setup for a multi-label text classification pipeline using DistilBERT. Include: 1. All necessary dependencies for deep learning (PyTorch, Transformers), NLP (NLTK), visualization (matplotlib, seaborn), and statistical analysis (scipy, statsmodels) 2. A seed-setting function that ensures reproducibility across NumPy, PyTorch, and CUDA 3. Support for both Google Colab and local environments 4. Compatibility with iterative-stratification for multi-label data splitting

2. Stage 2: Data Loading & Exploration

Notebook Reference: Cells 13, 15, 17, 19

Purpose: Understand the data structure, label distribution, and characteristics of the AI news headline dataset before preprocessing.

AI Prompt:

Create a comprehensive data loading and exploratory analysis module for a multi-label news headline dataset (Dataset A). Include: 1. Flexible file path detection for multiple environments (Colab, local, API) 2. A label parsing function that handles semicolon-separated multi-label strings 3. Statistical analysis of label distribution (frequency counts, co-occurrence patterns) 4. Visualization of label frequencies using both bar charts and pie charts 5. Display labels per sample distribution to understand multi-label density

3. Stage 3: Data Preprocessing

Notebook Reference: Cells 23, 25, 28

Purpose: *Clean and normalize text data while addressing class imbalance through rare label handling.*

AI Prompt:

Implement a text preprocessing pipeline for news headlines with the following components: 1. A `clean_text()` function that performs: - Lowercasing - URL and HTML tag removal - Whitespace normalization - Preservation of semantic content (no aggressive stemming) 2. Rare label handling with configurable threshold (default: 10 occurrences) 3. Option to merge rare labels into an "Other" category 4. Create bidirectional label-to-index mappings (`label2id`, `id2label`) Output should be model-ready while preserving the linguistic characteristics needed for transformer-based classification.

4. Stage 4: Multi-Label Encoding & Splitting

Notebook Reference: Cells 32, 34, 36

Purpose: *Convert multi-label annotations to multi-hot encoded vectors and create balanced data splits that preserve label distribution.*

AI Prompt:

Design a multi-label encoding and dataset splitting strategy with: 1. A `labels_to_multihot()` function converting label lists to binary vectors 2. Multi-label stratified sampling using iterative-stratification (`MultilabelStratifiedShuffleSplit`) 3. Train/Val/Test split ratios of 70%/10%/20% 4. Fallback to random splitting if iterative-stratification is unavailable 5. Validation of label distribution consistency across all splits 6. Visualization comparing label prevalence across train/val/test sets

5. Stage 5: Baseline Model (TF-IDF + Logistic Regression)

Notebook Reference: Cells 40, 42, 44, 47

Purpose: *Establish a fast, interpretable baseline to benchmark against more complex transformer models.*

AI Prompt:

Build an interpretable baseline multi-label classifier using: 1. TF-IDF vectorization with: - max_features=10000 - ngram_range=(1,2) for unigrams and bigrams - min_df=2, max_df=0.95 for frequency filtering 2. OneVsRestClassifier wrapping LogisticRegression with: - class_weight="balanced" for handling imbalance - liblinear solver for efficiency 3. Probability calibration using sigmoid on decision function scores 4. Comprehensive evaluation metrics: - Micro-F1 and Macro-F1 scores - Per-label precision, recall, F1, and support - Threshold optimization capability

6. Stage 6: DistilBERT Deep Learning Model

Notebook Reference: Cells 51, 53, 57, 59, 61

Purpose: Leverage pre-trained transformer representations for richer semantic understanding beyond bag-of-words approaches.

AI Prompt:

Implement a DistilBERT-based multi-label text classifier with 1. MultiLabelDataset class for PyTorch DataLoader compatibility: - Tokenization with configurable max_length (default: 128) - Attention mask handling - multi-hot label encoding 2. DistilBertMultiLabelClassifier architecture: - DistilBERT base (768-dim [CLS] representation) - Dropout layer (p=0.3) for regularization - Linear classification head → num_labels outputs 3. MultiLabelTrainer class with: - BCEWithLogitsLoss for multi-label training - AdamW optimizer with linear warmup scheduler - Training loop with progress tracking - Validation metrics: micro-F1, macro-F1 per epoch - Model checkpointing (save best model) 4. Hyperparameters: batch_size=16, lr=2e-5, epochs=3-5

7. Stage 7: Model Evaluation & Comparison

Notebook Reference: Cells 65, 68, 71

Purpose: Systematically compare baseline and deep learning models to understand their relative strengths and limitations.

AI Prompt:

Create a comprehensive model evaluation and comparison framework: 1. evaluate_distilbert() function returning predictions, ground truth, and probabilities 2. Side-by-side comparison table including: - Micro-F1, Macro-F1, Weighted-F1, Samples-F1 - Both baseline and DistilBERT results 3. Per-label confusion matrix visualization: - Select top K most frequent labels - Heatmap with annotation of TP, FP, FN, TN 4. Analysis of where each model excels/struggles 5. Recommendations for model selection based on use case

8. Stage 8: Interpretability Analysis

Notebook Reference: Cells 75, 77, 80

Purpose: *Provide human-interpretable insights into model decision-making and identify systematic prediction patterns.*

AI Prompt:

Develop interpretability tools for the multi-label classifier: 1. Extract high-weight TF-IDF features per label: - Top positive coefficients (indicative keywords) - Top negative coefficients (counter-indicative keywords) - Format as readable DataFrame 2. Error analysis function `find_high_confidence_errors()`: - Identify false positives with high probability - Identify false negatives with low probability - Return original text for qualitative inspection 3. Display keyword patterns for top 5 labels 4. Analyze why specific predictions failed (semantic ambiguity, label overlap)

9. Stage 9: Topic Modeling (NMF)

Notebook Reference: Cells 84, 86, 88

Purpose: *Uncover latent thematic structures to complement supervised classification with unsupervised patterns.*

AI Prompt:

Implement unsupervised topic discovery using Non-negative Matrix Factorization: 1. TF-IDF vectorization for topic modeling: - `max_features=5000` - `ngram_range=(1,2)` - `min_df=3`, `max_df=0.95` 2. NMF with 10 topics (`n_components=10`) 3. `display_topics()` function showing top keywords per topic 4. Assign dominant topic to each document 5. Visualize topic distribution: - Bar chart of topic frequencies - Topic confidence scores 6. Compare latent topics with supervised behavioral labels

10. Stage 10: Dataset C Analysis (LLM Comparison)

Notebook Reference: Cells 92, 94, 96, 99, 102, 107, 110, 113, 116

Purpose: *Assess how different LLMs frame AI's behavioral impact and identify systematic differences in their narrative patterns.*

AI Prompt:

Apply the trained classifier to Dataset C (LLM-generated text) for cross-model comparison: 1. Load Dataset C containing outputs from multiple LLMs (Mistral, Qwen, Llama) 2. Generate predictions using the trained baseline model 3. Analyze label distribution by LLM: - Per-LLM label frequency counts - Normalized proportions for comparison 4. Create visualizations: - Heatmap comparing label distributions across LLMs - Stacked bar chart for cluster-level analysis 5. Statistical testing: - Chi-Square test for distribution differences - Jensen-Shannon divergence for similarity measurement 6. Prompt type effect analysis: - Classify prompts by keywords (teacher, student, creative, etc.) - Analyze how prompt framing affects behavioral outputs 7. Integrated visualization dashboard (2x2 grid)

11. Stage 11: Quantitative Financial Analysis

Notebook Reference: Cells 122, 124, 126, 128, 133, 135, 138, 141, 144

Purpose: *Investigate whether AI news behavioral framing has predictive power for market volatility and sector performance.*

AI Prompt:

Build a quantitative analysis layer linking news behavioral labels to market dynamics: 1. Define AI-related stock/ETF tickers (NVDA, GOOGL, MSFT, META, etc.) 2. Create industry exposure matrix mapping labels to sectors 3. Aggregate news by date generating daily label time series 4. Generate/acquire market data: - Daily returns - Volatility measures 5. GARCH (1,1) volatility modeling: - Fit conditional volatility - Annualized volatility estimates 6. Correlation analysis: - News label proportions vs. market returns - News sentiment vs. volatility - Heatmap visualization 7. Granger causality testing: - Test if news labels predict market volatility - Report significant causal relationships 8. Industry exposure analysis: - Compute daily exposure scores - Link coverage to market correlations 9. Create visualization dashboard: - News volume vs. market volatility - Correlation heatmaps - Granger causality significance